

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B.Tech Examination (CE)
December 2012

CE209- Design & Analysis of Algorithm

Date: 08.11.2012, Thursday

Time: 01:30 p.m to 4:30 p.m

Maximum Marks:70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

- Q-1**
- (a) What is the difference between backtracking and branch and bound ? 2
 - (b) Define the term complexity of an algorithm and explain worst-case and average case analysis of an algorithm. 2
 - (c) What is Asymptotic notation? Define O , Ω , θ notations. 3

- Q-2**
- (a) Write insertion sort algorithm and analyze the running time of the algorithm in all three cases. 5
 - (b) Explain Greedy choice property and optimal substructure property of greedy algorithm. 4
 - (c) Sort the following sequence using merge sort. {1, 4, 5, 15, 4, 10, 7, 18, 5, 14, 35, 49, 43, 21}. Find the Asymptotic cost of division, conquer and combine. 5

OR

- (b) What is articulation point and Biconnected components of a graph? Explain with the help of an example. 4
- (c) Define algorithm and mention the parameters for selecting suitable algorithm among many candidate algorithms. Justify parameter with suitable example. 5

Q-3 Attempt Any Two. 14

- (a) Write an algorithm for quick sort and show the tracing of the algorithm using following data.
15 14 13 16 20 21 3 9 10 30
- (b) Solve the following fractional Knapsack Problem using Greedy Algorithm. There are five items whose weights and values are given in following arrays.
Weight $w[] = \{ 1, 3, 5, 8, 9 \}$ Value $v[] = \{ 10, 6, 10, 21, 24 \}$
- (c) Solve the following recurrence where n is exactly power of 2.
$$T(n) = \begin{cases} 1 & \text{if } n=1 \\ 4T(n/2) + n \log n & \text{otherwise} \end{cases}$$

SECTION-II

- Q-4**
- (a) Arrange the following terms in the ascending order of their running time. 2
 $O(N \log N)$, $O(1)$, $O(N)$, $O(\log N)$, $O(2^N)$, $O(N^2)$, $O(N^3)$
 - (b) What is recursion? A recursive procedure should have two properties. What are they? 2
 - (c) What are the differences between divide & conquer technique and dynamic programming? 3

Q-5

- (a) Find minimum Spanning Tree for graph shown in Fig 1 using Prim's algorithm.

6

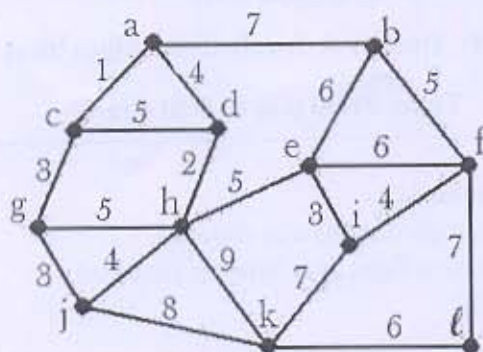


Fig 1

- (b) Solve the following recurrence equation:

4

$$f(n) = \begin{cases} n & n=0,1 \\ f_{n-1} + f_{n-2} & \text{otherwise} \end{cases}$$

- (c) Explain the difference between following graph representations.

4

1. Adjacency Matrix
2. Adjacency List

OR

- (b) Explain how to find out Longest Common Subsequence of two strings using Dynamic Programming method. Find any one Longest Common Subsequence of given two strings using Dynamic Programming. S1=abbacdcba S2=bedbbcaac

4

- (c) Solve using recurrence tree method:

4

$$T(n) = 3T(2n/3) + 2T(n/3) + C, \text{ where } C \text{ is a constant.}$$

Q-6 Attempt Any Two.

14

- (a) Find Optimum Parentheses order of Matrix Multiplication for Dimension vector [2,4,5,6,3].
- (b) Let $S = \{a, b, c, d, e, f, g\}$ be a collection of objects with benefit-weight values as follows :
 $a : (12,4)$, $b : (10,6)$, $c : (8,5)$, $d : (11,7)$, $e : (14,3)$, $f : (7,1)$, $g : (9,6)$. What is an optimal solution to 0/1 knapsack problem using dynamic programming assuming the sack can hold objects with total weight 18?
- (c) Find Solution of n queen Problem using Back tracking with $n=5$.